Analytical Questions - Unit 1

1. Ceaser Cipher

Cipher text : PHHW PH DIWHU WKH WRJD SDUWB

Find the Key

Try Common Ceaser Shift

* PHHW → MEET using shift 3

∴ Key = 3

plain text : MEET ME AFTER THE TOGA PARTY

* Ceaser Cipher shifts every letter by same amount.

2. Ceaser Cipher (Key = 11)

Plain text : SECURITY ; Key = 11

$$C = (P+11) \bmod 26$$

Plain	Value	C	Cipher
S	18	3	D
E	4	15	P
C	2	13	N
U	20	5	F
R	17	2	C
I	8	19	T
T	19	4	E
Y	24	9	J

Ciphertext : DPNFCTEJ

3, Why Ceaser Cipher is Vulnerable?

only 26 possible Keys exist

An attackers simply tries every shift

Ex :-

Key 0

Key 1

Key 2

⋮

Key 25

Time Complexity : $O(26) \approx O(1)$

* Since 26 is Constant, brute force is extremly easy.

4, Monoalphabetic Cipher :

Possible Keys : $26! \approx 4 \times 10^{26}$

Although the key space is huge, English letters occur with predictable frequencies.

Example :- E → Most Common

T → Second

A → third

O → fourth

Hence it is Vulnerable to frequency Analysis.

5, Ciphertext.

GSRH RH Z HVXIVG

Atbash Cipher

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U					
Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F					

W X Y Z

D C B A

⇒ plain tex :- THIS IS A SECRET

6, Frequency Analysis

Ciphertext letter X appears most often.

English most frequent letter is E

Intial assumption : X → E

1, Replace X with E

2, Examine common words.

3, Look for patterns like THE, AND, ING, TION

4, Adjust Substitutions until readable english appers.

7, Monoalphabetic :

E → always X

Polyalphabetic :

E → X , E → P, E → M , E → T

The same plaintext letter encrypts to different ciphertext letters depending on the key.

8, Plain text : CRYPTOGRAPHY

Key : KEYKEYKEYKEY

Plain	P	Key	K	Cipher
C	2	K	10	M
R	17	E	4	V
Y	24	Y	24	W
P	15	K	10	Z
T	19	E	4	X
O	14	Y	24	M
G	6	K	10	Q
R	17	E	4	V
A	0	Y	24	Y
P	15	K	10	Z
H	7	E	4	L
Y	24	Y	24	W

Ciphertext : MVWZXMQVYZLW

9, Plaintext :

Cipher text : LXFOPVEFRNHR

Key : LEMONLEMONLE

Subtract Key Values

Result : ATTACKATDAWN

10, Ex:-

Key : LEMON

LEMONLEMONLEMON.....

The repetation Creates pattern

1, Search for repeated ciphertext groups.

2, Measure distances btw repeats.

3, Compute common factors.

4, Estimate Keyword length.

5, Break into Ceaser ciphers.

11,

* If the vigenere Key is random, Same length plaintext and use only once, it becomes one-Time Pa

* Security: Perfectly Secure & theorotically unbreak

12,

Monoalphabetic	Polyalphabetic
* one alphabet	* Multiple alphabets
* Low entropy	* High entropy
* Low Confusion	* High confusion
* Less Secure	* More Secure

13, * Letters are assigned values A=0, B=1 ---, Z=25

* Encryption : C = (P+K) mod 26

* Decryption $P = (C-K) \bmod 26$

* Mod 26 Keeps values within the alphabet.

14, Key matrix : $\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$

plaintext : HI

* H = 7 , I = 8

$$\Rightarrow \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} \Rightarrow \begin{bmatrix} 23 \\ 61 \end{bmatrix}$$

* Mod 26 ⇒ (23, 9)

∴ Ciphertext = XJ

15, matrix ⇒ $\begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}$

* Determinant = 0
* No modular inverse exists.
* Encryption | Decryption fails because the matrix is not invertible.

16, * The key matrix need an inverse for decryption.
* This is possible only iff gcd (det, 26) = 1
* Otherwise, decryption impossible.

17,
* Ceaser : Encrypts one letter at a time.
* Hill : Encrypts blocks using matrix multiplication.
* Hill provides better security and diffusion.

18, * Indicates a polyalphabetic Substitution Cipher.
* The Same plaintext maps to different Ciphertext letters.

19, Example Key :

$A \leftrightarrow Q$, $B \leftrightarrow W$, $C \leftrightarrow E$, $D \rightarrow R$ --- $Z \rightarrow M$,

* Share the Key Securely.

* Never send the Key with Ciphertext.

20, * Yes, Identical plaintext blocks produce identical Cipher-text blocks when the Same Key is used

* This Can reveal patterns, reducing Security

21, Ceaser Cipher

$$C = (7P+3) \bmod 26$$

* No, it is not a simple shift cipher.

* It is an Affine Cipher, as it uses multiplication & addition.

22, * The attacker Knows Some plaintext and its cipher

* By Comparing them, the substitution Key can be discovered.

23, * Better Confusion:

Same plaintext letter Can encrypt differently.

* Limited diffusion:

One plaintext letter affects only one Ciphertext letter.

24, * 'THE' is Common English word.

* The attacker Searches for repeated 3-letter patterns

* This helps recover other letter Substitutions

25, 1, Ceaser Cipher - Very Weak

2, Monoalphabetic cipher - Large key Space but Vulnerable to frequency analysis.

3, Vigenere cipher - More Secure

4, Hill cipher - Strongest among these.

26, Cipher text : RRMTEEIATETMLDYAE

* It is not correct for plaintext RETREATIMMEDIATELY using 3 rails.

* Reason: Zig-Zag was applied incorrectly.

27, * plaintext is written row-wise into 4 Columns.

* If last row is incomplete, padding characters may added.

* Ciphertext is obtained by reading Column-wise.

28, * Repeated letters cause column numbering ambiguity

* A fixed rule is required.

* Otherwise encryption & decryption become inconsistent.

29, * Plaintext : SECRETMISSION

* Ciphertext : SCEIINERTSSMO

* Ciphertext is incorrect.

* The zig-zag placement for 2-rail fence was not followed properly.

30, Ciphertext length : 24

Possible matrix sizes :

⇒ 4x6

⇒ 6x4

* Since 24 is exactly divisible by 4&6 :

⇒ No padding required.

* Double transposition applies two successive colu
permutations.

31,

KEY : ZEBRA

* Arrange Alphabetically : A, B, E, R, Z
* column order : 5, 3, 2, 4, 1
* Read columns in this order to recover the plaintext.

32,

* Transposition only changes the positions of let
* Letter frequencies remain unchanged, so frequency analysis cannot reveal the key.

33,

* ~~this~~ Key 1 → (2, 1, 3, 4)

Key 2 → (2, 1, 3, 4)

⇒ Same permatation

⇒ Same ciphertext.

34, Try → R = 2 ✓
R = 3 ✓
R = 4 ✓
R = 5 ✓

Readable plaintext ⇒ Correct rails.

35, Key length = 5

$5! = 120$ ⇒ Total Keys $= 120 \times 120$
$= 14400$

36, plaintext : HELLO

HELLO + X ⇒ HELLOX

padding = 1 character

37, Length = 17

17 is prime ⇒ Possible :
17×1 ; 1×17

Other sizes require padding

38, Correct

Row → Column

Wrong

Column → Row

Result = Incorrect plaintext

39, Rail 1 : C P

Rail 2 : R A T

Rail 3 : P O

First letter : C , last letter = O

∴ outer rails remains unchanged

40, Plaintext : 23, Key - 6

Rows $\Rightarrow \frac{23}{6} \Rightarrow 4$

Cells $\Rightarrow$ $6 \times 4 = 24$

Empty cells $\Rightarrow$ $24 - 23$
$= 1$

Padding required = 1

41, Key length = 5

$5! = 120$ $\rightarrow$ Total = 120×120
$= 14400$

42, Step 1 : Try single Transposition
↓
Meaningful plaintext?
No
↓
Try Double Transposition

43, Key = BALLOON

Repeated letters
L, L
O, O

Column numbering becomes ambiguous.
· use left → Right numbering.

44, ABCDE ABCDE ABCDE
↓
Period = 5

More likely Columnar Transposition than Rail fen

45, Ciphertext len = N

Rows have nearly equal letters

↓

Likely rails = 3 or 4

∴ Equal distribution indicates balanced rail allocation.

46, Key1 len = 6

Key2 len = 3

$6/3 = 2$

∴ Repeating patterns may occur, reducing security

47, plaintext = 22 , Key length = 6

Rows ⇒ $22/6 = 4$

Cells $= 6 \times 4 = 24$

padding $= 24 - 22 = 2$

48, Key length = 6

possible Keys = $6!$ ⇒ $6 \times 5 \times 4 \times 3 \times 2 \times 1$

$= 720$

49, plaintext : CRYPTOGRAPHY

Try : 2 rails ✗

3 rails ✓

∴ Ciphertext corresponds to 3-rail fence cipher.

50,

Numeric Key	Alphabetic Key
* Easy to use & order * No sorting * Less ambiguity	* Easy to remember * Must convert to numbers * May cause ambiguity.